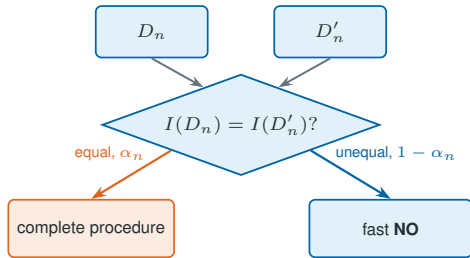


A Pair collision and complete branch



$$\alpha_n = \mathbb{P}\{I(D_n) = I(D'_n)\} = \mathbb{P}\{(D_n, D'_n) \in \mathcal{B}_n(I)\}$$

Equality is the black-hole region of this invariant-first comparison.

B Database birthday scale

$$\mathbb{E}C_M = \binom{M}{2} \alpha_n$$

expected collisions = expected complete-procedure calls

$$\begin{array}{lll}
 M^2 \alpha_n \rightarrow 0 & M^2 \alpha_n \rightarrow \lambda & M^2 \alpha_n \rightarrow \infty \\
 \text{collision-free w.h.p.} & \text{birthday window} & \text{collision w.h.p.}
 \end{array}$$

The diagram shows three asymptotic regimes for the parameter $M^2 \alpha_n$. The first regime, $M^2 \alpha_n \rightarrow 0$, is labeled 'collision-free w.h.p.'. The second regime, $0 < \lambda < \infty$, is labeled 'birthday window'. The third regime, $M^2 \alpha_n \rightarrow \infty$, is labeled 'collision w.h.p.'.

$$M_B(n) = \alpha_n^{-1/2} = \exp(H_2/2)$$

The three database ranges are determined by $M^2 \alpha_n$.